

GAZE CONTINGENCY SEARCH STUDY

Source Version 0.1, September 16, 2026; meeting annotations retained

Formatted September 23, 2026

Setting Seated, rotating mixed-reality object search

Hardware HTC Vive Focus Vision

Status Protocol proposal / researcher review

EXPERIMENTER RUN SHEET

Follow My Voice

SAY / ASK Participant-facing speech is set in dark upright type.

ACTION / RESEARCHER *Experimenter-only instructions are italic and muted gold-brown. Do not read them aloud.*

Participant ID _____ **Run** _____

Date _____ **Researcher** _____

Build _____ **Voice order** _____

Preferred voice ID _____ **Survey version** _____

RESEARCHER ONLY *Two counterbalanced voice blocks, participant-preferred and self-similar. Both use gaze-contingent warmer/colder guidance. Planned trial start is the same fixed front direction. Say “first block” and “second block”; do not suggest that either voice should improve performance.*

Before participant arrives

- *Sanitize and test the Head Mounted Display (HMD) and controllers.*
- *Open Unity and Qualtrics.*
- *Enter coded ID and assigned condition.*
- *Prepare consent, baseline survey, both block surveys, final interview and session/event record. Use one defined TLX collection route; see the beta note in Section 7.*

1 Welcome and consent

SAY “Thank you for coming. You will stay seated and turn to find virtual objects by their color and shape. A voice assistant will provide hints. You may pause, take a break or stop at any time without penalty.”

SAY “We will ask you to record a short voice sample to create a synthetic version of your voice. We also record eye-gaze and task data under a study code. Please read the consent form for what is collected, where it is processed and how it is handled.”

ACTION *Obtain consent.*

2 Initial Qualtrics and Voice Setup

PROTOCOL PROPOSAL / SETUP REQUIREMENTS

This section specifies the requested survey and voice-selection flow. Verify the matching Qualtrics version and application workflow before participant use; this run-sheet edit does not establish that those features are implemented. Agent gender and accent preferences are participant choices, not inferred from the participant's identity, appearance or voice.

SAY "Please complete the background and current-comfort questions before we begin. Use only the study code already entered. Do not add your name or contact details. You may skip optional questions. We will also ask what kind of assistant voice you would prefer. There are no right or wrong preferences."

ACTION / INITIAL QUALTRICS *Confirm the coded participant and run IDs. Include the baseline demographics and current-comfort items, followed by the separate questions below. Keep skipped responses distinct from "no preference."*

SURVEY ITEM / OWN ACCENT "What do you perceive your own accent to be?"

RESPONSE FORMAT *Optional free text; also allow "Not sure" and "Prefer not to answer." Do not infer an accent from nationality or first language.*

SURVEY ITEM / AGENT GENDER PREFERENCE "What gender presentation would you prefer for the agent's voice?"

RESPONSE FORMAT / PROPOSAL *Single choice: feminine-sounding; masculine-sounding; androgynous or gender-neutral-sounding; no preference; another preference (optional description); prefer not to answer. These describe a voice preference, not the participant's gender. Confirm the labels against the available voice library before publishing the survey.*

SURVEY ITEM / AGENT ACCENT PREFERENCE "What accent would you prefer the agent's voice to have?"

RESPONSE FORMAT *Offer the accents actually represented in the candidate library, plus "No preference," "Another accent" (optional description), "Not sure" and "Prefer not to answer." A response of "an accent like mine" may be retained explicitly if offered; do not assume it from the own-accent answer.*

ACTION / EARLY RECORDING *After consent and these initial answers, collect the voice sample early so preparation can run during the remaining setup. Use the appendix passage and the approved recording workflow; record success, failure and retries under the study code. Do not start task trials while either required voice is unread.*

SAY "We will now record the short voice sample described in the consent form. After the tone, please read the displayed passage in your usual speaking voice."

ACTION / CANDIDATE PREPARATION *Use the stated agent-gender and accent preferences to prepare the available candidate set. Retain the planned five-to-six-candidate evaluation where feasible. Save the requested preferences separately from the exact candidates offered and their library labels. Do not automatically match agent gender to participant gender or silently substitute an unavailable accent. Candidate-set and presentation-order rules must be consistent across participants and recorded.*

SAY “You will hear several possible assistant voices. Please listen to each example and rate how much you like it. You can choose the voice you prefer after hearing the examples. You do not have to choose a voice that sounds like you.”

ACTION / PREVIEWS AND RATINGS *Present every candidate using the same preview text and comparable playback level. Save each exact voice ID, presentation order and individual liking score using the study’s finalized 100-point rating format. Record the selected preferred voice; if ranks are also collected, retain them separately from ratings. Do not choose on the participant’s behalf when there is a tie or no preference.*

ASK “Which voice would you prefer to use for the task? Does it fit the gender presentation and accent you wanted? It is fine if none of these voices fits your preference.”

ACTION / MISMATCH OR NO PREFERENCE *Record the participant’s response. If the requested option is unavailable or none is acceptable, explain the available alternatives and let the participant choose an acceptable alternative. Preserve the original preference, offered alternatives and final choice. If no choice can be made, use only a predefined, documented allocation rule or pause setup for the researcher; do not silently assign a preferred voice.*

ACTION / SELF-SIMILAR VOICE CHECK *When the synthetic voice is ready, check audibility and clipping with the participant. Record any playback failure or retry. Collect the planned comparable liking rating for the self-similar voice at the predefined setup time. Keep own-voice similarity and accent similarity as separate seven-point items; use the finalized survey anchors and save instrument version and exact voice IDs. Standardize and log this pre-task voice exposure.*

SURVEY ITEMS / SEPARATE RATINGS “The agent’s voice is similar to mine.” “The agent’s accent is similar to mine.”

ASK / AUDIO CHECK “Can you hear the voice clearly and at a comfortable volume?”

ACTION / SETUP COMPLETE *Save survey responses, requested gender/accent preferences, all candidate scores, final preferred voice ID, self-similar voice ID and readiness status under the same participant/run identifiers. Confirm the assigned counterbalanced block order independently of voice preferences. Use neutral “first block” and “second block” wording. Final rating endpoints, candidate ordering, tie/no-preference allocation and similarity-item timing remain protocol decisions to freeze before participant use.*

3 Fit and calibrate the headset

SAY “Remain seated in this chair. You may turn the chair, your body and your head to look around, but do not stand or walk. Tell me immediately if you feel dizzy, nauseated, strained, uncomfortable or unsteady.”

ACTION *Fit equipment and verify tracking, perform the automated eye-calibration protocol.*

ASK “Can you see clearly? Does the equipment feel comfortable? Do you feel well enough to continue?”

4. Pre-Block Activity

SAY “First, I will explain the task. Your task is simply to locate objects that are around you. The objects that you’ll need to find are specified by an agent. For example, the agent might say: “Locate the blue sphere”. Once you find the object, point the controller ray at it to highlight it, then press the trigger to select it – this completes the round. You will then continue this for other objects the agent specifies.

ACTION *Voice recording and preferred-voice choice were completed in Section 2. Confirm both voices are ready and the saved preferred voice is correct; do not repeat enrollment here unless recovery is necessary and logged.*

Meeting updates

RESEARCHER NOTES

The following requirements were added to the supplied run sheet. They remain proposals/open decisions; the GitHub backlog owns implementation and validation.

MEETING UPDATES / PROPOSED MEASURES

- Preferred accent
- Gender perception
- Superseded by Section 2: collect the participant's preferred agent-gender presentation; do not infer or automatically match it to participant gender.
- Ask for 100 point rating on each voice (including self-similar)
- ("The agent's voice is similar to mine.", The agent's accent is similar to mine.) (1-7), Likert Scale
- Log time to first gaze
- Log time to first controller raycast/object intersect
- Log overall trial time (includes final selection trigger)
- After action review tool (Participant Trial Replay) (Controller + Eye Gaze), is trigger held down (1/0) (60 HZ) + object position
- Total Gross Rotation + Head Movement (Y + X axis + Z), yaw/pitch/ per round'
- Drewes Gaze vs locomotion paper
- [10] J. Drewes, S. Feder, and W. Einhauser. Gaze during locomotion in virtual reality and the real world. *Frontiers in Neuroscience*, 15:656913, 2021. doi: 10.3389/fnins.2021.656913 3, 6 D. C. Niehorster, M. Nystrom, R. S. Hessels, J. S. Benjamins, R. Andersson, and I. T. C. Hooge. The fundamentals of eye tracking, part 7: 8Online Submission ID: 2013 Determining data quality. *Behavior Research Methods*, 58:183, 2026. doi: 10.3758/s13428-026-03039-4 3, 7
- use colors and italics to actions to perform without speaking
- update start of protocol, include mention of self-similar voice clone use in the experiment

5. Familiarization

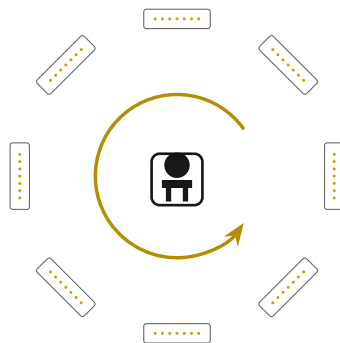
ACTION *At the 360 search start checkpoint, seat the participant at the intended center facing the chosen room-front reference. Release the trigger, then press Trigger / Enter to center and begin. Record the reference; do not recenter to wherever the participant happens to face later.*

SAY “Objects will appear around you, you can turn in your chair in all 360 degrees. Find the object with the announced color and shape. Turn while staying seated, point the controller ray at the matching object to highlight it, then press the trigger to select it. Looking alone does not select an object. You do not need to count rounds.”

MEETING NOTE

The source requests a simple seated 360-degree illustration. The schematic below adds that visual without specifying a selection method.

Objects can appear around you



Remain seated. Turn to look around.

Schematic, not to scale. Illustrates seated rotation only.

SAY “The next rounds are practice, and are meant to simply get you familiar with the task.”

ACTION *Allow the user to complete the object selection practice rounds.*

ASK “Do you have any questions about the controls or task?”

SAY “When you are ready, we will begin the first task block.”

6 First task block

INTERNAL SETTINGS *Practice | Real | 30 rounds*

RESEARCHER SETTINGS *Confirm the saved first voice and planned trial schedule. Keep object layout, selection method, guidance policy and the finalized prompt wording constant across voices.*

SAY “We will now begin the first task block. Find each target as quickly and accurately as you comfortably can. Stay seated and use the same controller ray and trigger selection you practiced. Tell me if you need to pause.”

ACTION *Start block; monitor safety.*

AT COMPLETION SAY “That task block is complete. Please stop and lower your hands comfortably.”

SAY “These questions ask about your workload, your experience of the activity and the agent. There are no right answers. Please answer based on the block you just finished.”

ACTION *Administer Survey Block #1.*

SAY “You may take a short break of up to 5 minutes. Tell me when you are ready to continue.”

BEFORE CONTINUING ASK “Do you feel comfortable and well enough to continue?”

Second Task Block

INTERNAL SETTINGS *| Assigned condition | 30 valid attempts*

INTERNAL SETTINGS *READ ONLY THE ASSIGNED CONDITION SCRIPT:*

SAY “We will now begin the second task block. The task and selection method are the same. Find each target as quickly and accurately as you comfortably can, and tell me if you need to pause.”

ACTION *Start block; monitor safety.*

AT COMPLETION SAY “The search task is complete. Please stop turning. We will finish the questions about this block, then I will help you remove the headset.”

SAY “These questions ask about your workload, your experience of the activity and the agent. There are no right answers. Please answer based on the block you just finished.”

ACTION *Administer Survey Block #2.*

9 Final Qualtrics

SAY “Please complete the final questionnaire. It asks about your experiences with the two voices. It is fine to have no preference. We are interested in what you experienced, including anything distracting, uncomfortable or unhelpful.”

SAY “Some follow-up questions may appear depending on your answers and the timing condition you experienced. Answer based on your own experience and tell me when you have submitted it.”

EXPLORATORY FOLLOW-UP *After open responses, ask “Did either voice feel like your own thoughts, like another person helping, or neither?” Record this as an exploratory report, not proof of an internal-dialogue mechanism.*

ACTION *Administer Post-Study Survey Block*

10 Debrief and finish

SAY “Thank you. This study compares participant-preferred and self-similar voices during gaze-guided object search. Both voices provide gaze-contingent assistance. We are examining task performance and how participants experience the activity and assistant.”

SAY “Your data are stored under a coded participant ID. Please avoid discussing the timing condition or study details with people who may participate later, because knowing them in advance could affect their behavior.”

ASK “Do you have any questions about the study or your experience?”

SAY “Thank you for participating. We are finished.”

Immediate stop or pause criteria

PAUSE OR STOP

Participant request, dizziness, nausea, headache, eye strain, fatigue, pain, distress, unsafe movement or equipment failure. Help the participant remain seated and remove the headset when needed.

ACTION *Pause through the verified researcher control; the beta supports P on a connected keyboard and Trigger / Enter at its resume checkpoint. App focus loss can pause or technically stop a session. Log time, trial, reason and outcome. Resume only when the participant wants to continue and tracking/audio/alignment are valid; otherwise stop and preserve partial data.*

Final researcher check

- *Confirm both block surveys and final interview/symptom records, or record why incomplete. Match participant, run, block and voice identifiers.*
- *Secure data and sanitize equipment.*

Appendix Current recording passage

CURRENT RECORDING PASSAGE

“Locate the red sphere. Look toward the upper shelf. Move your gaze slowly to the right. Check the blue cube near the center. Compare each object’s color and shape. Scan the lower row from left to right. Look beside the purple cylinder. Point the controller ray at the target and check its highlight. Press the trigger to select the matching object. Take a short pause. Get ready for the next round.”